



*Energy Harbor Nuclear Corp
Perry Nuclear Power Plant
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Perry, Ohio 44081*

Rod L. Penfield
Site Vice President, Perry Nuclear

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June 27, 2021
L-21-200

10 CFR 50.73(a)(2)(i)(B)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2021-001, "Division 3 EDG Inoperability Resulting in an Operation or Condition Prohibited by Technical Specifications." There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Glendon Burnham, Manager – Regulatory Compliance, at (440) 280-7538.

Sincerely,

A handwritten signature in black ink, appearing to read "Rod L. Penfield", written over a horizontal line.

Rod L. Penfield

Enclosure:
LER 2021-001

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III Regional Administrator

Enclosure
L-21-200

LER 2021-001



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	2. Docket Number	3. Page
Perry Nuclear Power Plant	05000 00440	1 OF 2

4. Title
Division 3 EDG Inoperability Resulting in an Operation or Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
06	01	2021	2021	- 001 -	00	07	28	2021		05000
									Facility Name	Docket Number
										05000

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Hali Jenkins – Regulatory Compliance	440-280-6378

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	EK	RG	B093	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 1, 2021, with the Perry Nuclear Power Plant operating at 100 percent power an operation or condition prohibited by Technical Specifications occurred. During the monthly Division 3 Diesel Generator (DG) run on June 1, 2021, erratic output voltage was noted for approximately 45 seconds after start. At the time operations had reasonable expectation of operability and requested a Follow-up Operability Determination (FOD). Results from the Division 3 DG FOD could not support continued Operability of the Division 3 DG. The Division 3 DG was declared inoperable and unavailable on June 3, 2021 at 18:15.

The most plausible cause for the erratic output voltage falling below the allowable band during startup is an internal infant component mortality failure of the voltage regulator. The component will be sent for failure analysis.

On June 1, 2021 at 10:15, when erratic output voltage was first noted for the Division 3 Diesel Generator, entry into TS 3.8.1 Condition B and the accompanying actions did not occur. Surveillance requirement 3.8.1.1 was not performed as required within one hour and subsequent action TS 3.8.1 Condition F was also not met to be in Mode 3 in 12 hours. Since completion times were not met for TS 3.8.1 Condition F, this issue is reportable under 50.73(a)(2)(i)(B) as an operation or condition prohibited by the plant's technical specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant	05000-	2021	- 001	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

EVENT DESCRIPTION

During the monthly Division 3 Diesel Generator (DG) [EK] run on June 1, 2021, erratic output voltage was noted for approximately 45 seconds after start. Operations had reasonable expectation of operability and requested a Follow-up Operability Determination (FOD). Results from the Division 3 DG FOD could not support continued Operability of the Division 3 DG. The Division 3 DG was declared inoperable and unavailable on June 3, 2021 at 18:15. On June 4, 2021 the new voltage regulator [RG] was replaced and was tested satisfactory and the DG returned to operable status at 21:25.

A review of the Division 3 DG maintenance history revealed that the voltage regulator was replaced and passed the Post Maintenance Test (PMT) on May 15, 2021. No other failures of the Division 3 DG or the voltage regulator could be found during the previous three years.

On June 1, 2021 at 10:15, when erratic output voltage was first noted for the Division 3 Diesel Generator, entry into TS 3.8.1 Condition B and the accompanying actions did not occur. Surveillance requirement 3.8.1.1 was not performed as required within one hour and subsequent action TS 3.8.1 Condition F was also not met to be in Mode 3 in 12 hours. Since completion times were not met for TS 3.8.1 Condition F, this issue is reportable under 50.73(a)(2)(i)(B) as an operation or condition prohibited by the plant's technical specifications.

CAUSE

The most plausible cause for the erratic output voltage falling below the allowable band during startup is an internal infant component mortality failure of the voltage regulator. The component will be sent for failure analysis.

EVENT ANALYSIS

A Probabilistic Risk Assessment (PRA) evaluation was performed for the June 3, 2021 Division 3 Diesel Generator voltage regulator Inoperability event. Fire and other external hazards were qualitatively addressed to be of low risk. The analysis of this event results in delta CDF and delta LERF values that are below the acceptable thresholds of 1.0E-06/yr and 1.0E-07/yr, as discussed in Regulatory Guide 1.174. The risk of this event is therefore considered to be of very low safety significance in accordance with the Regulatory Guidance.

CORRECTIVE ACTIONS**Completed Actions**

On June 4, 2021 the voltage regulator was replaced and was tested satisfactory.

Scheduled Actions

A corrective action was created to send the defective voltage regulator for failure analysis testing. Further corrective action will be taken based on the results.

PREVIOUS SIMILAR EVENTS

No failures of the Division 3 DG or the voltage regulator could be found during the previous 3 years.

COMMITMENTS

None